CONCEPT CHECK 52.4

1. Describe human actions that could expand a species' range by changing its (a) dispersal or (b) biotic interactions.
2. **WHAT IF?** You suspect that deer are restricting the distribution of a tree species by preferentially eating the seedlings of the tree. How might you test this hypothesis?
3. **MAKE CONNECTIONS** Hawaiian silverswords underwent a remarkable adaptive radiation after their ancestor reached Hawaii, while the islands were still young (see Figure 25.23). Would you expect the cattle egret to undergo a similar adaptive radiation in the Americas (see Figure 52.19)? Explain.

For suggested answers, see Appendix A.

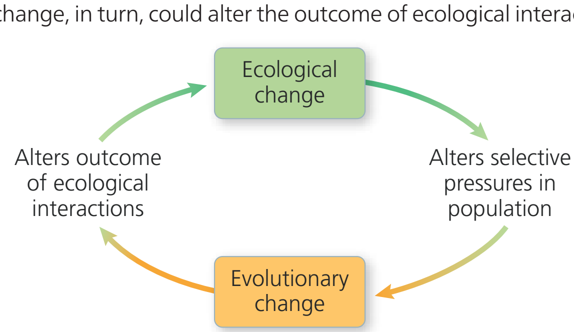
CONCEPT 52.5

Ecological change and evolution affect one another over long and short periods of time

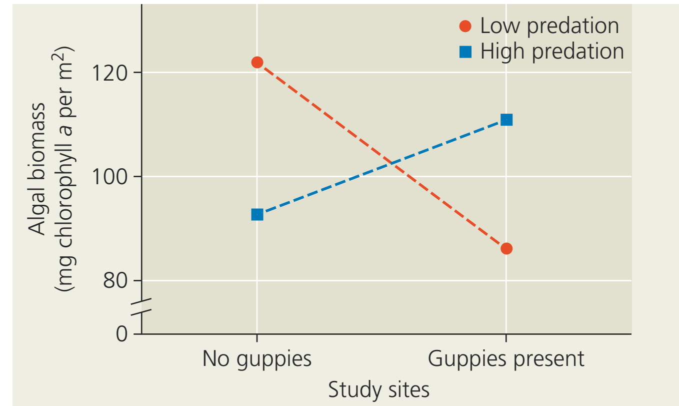
Biologists have long recognized that ecological interactions can cause evolutionary change, and vice versa **(Figure 52.24)**. The history of life includes many examples of these reciprocal effects occurring over long periods of time. Consider the origin and diversification of plants. As described in Concept 29.3, the evolutionary origin of plants altered the chemical cycling of carbon, leading to the removal of large quantities of carbon dioxide from the atmosphere. As the adaptive radiation of plants continued over time, the appearance of new plant species provided new habitats and new sources of food for insects and other animals. In turn, the availability of new habitats and new food sources stimulated bursts of speciation in animals, leading to further ecological changes. Here, as in many other such examples, ecological and evolutionary changes had ongoing and major effects upon one another.

The interplay between ecological and evolutionary change illustrated by the origin of plants occurred over millions of years. Reciprocal "eco-evolutionary" effects occurring over centuries or thousands of years are also well documented, as in the examples of the mosquitofish and the apple maggot fly

▼ **Figure 52.24 Reciprocal effects of ecological and evolutionary change.** An ecological change, such as the expansion of a predator's range, can alter the selective pressures faced by prey populations. This could cause evolutionary change, such as an increase in the frequency of a new defensive mechanism in a prey population; that change, in turn, could alter the outcome of ecological interactions.



▼ **Figure 52.25 Rapid evolution leading to rapid ecological change.** Dotted lines connect algal abundance at control sites (with no guppies) to algal abundance at nearby sites in the same streams inhabited by guppies that had evolved under either low or high levels of predation.



discussed in Concept 24.2. But are such joint effects common over even shorter periods of time? As we've seen in previous chapters, ecological change can cause evolutionary change over the course of a few years to decades; examples include beak length evolution in soapberry bugs (see Figure 22.13) and the formation of new sunflower species (see Figure 24.18).

Recent studies show that the causation can run both ways: Rapid evolution can also cause rapid ecological change. For example, Trinidadian guppy (*Poecilia reticulata*) populations evolve rapidly when predators are removed: Guppy color patterns, jaw morphology, and feeding preferences all change within a few years. In 2017, researchers showed that these rapid evolutionary changes affect the stream ecosystems in which the guppies live. For instance, guppies that had evolved under different levels of predation had contrasting effects on algal abundance **(Figure 52.25)**. Guppies that had evolved under low predation fed primarily on algae (thereby reducing algal abundance), while guppies that had evolved under high predation fed primarily on invertebrates (thereby tending to increase algal abundance because some invertebrates eat algae). As producers, algae are key components of the ecosystem: Other organisms depend on them for food, either directly (by eating them) or indirectly (by eating an organism that ate algae). Overall, this and other studies show that ecological change and evolution have the potential to exert rapid feedback effects on each other.

CONCEPT CHECK 52.5

1. Describe a scenario showing how ecological change and evolution can affect one another.
2. **MAKE CONNECTIONS** Fisheries target older, larger cod fish, causing cod that reproduce at a younger age and smaller size to be favored by natural selection. Younger, smaller cod have fewer offspring than do older cod. Predict how evolution in response to fishing would affect the ability of a cod population to recover from overfishing. What other reciprocal eco-evolutionary effects might occur? (See Concept 23.3.)

For suggested answers, see Appendix A.